

PERSON IDENTIFICATION SYSTEM

PROJECT TEAM

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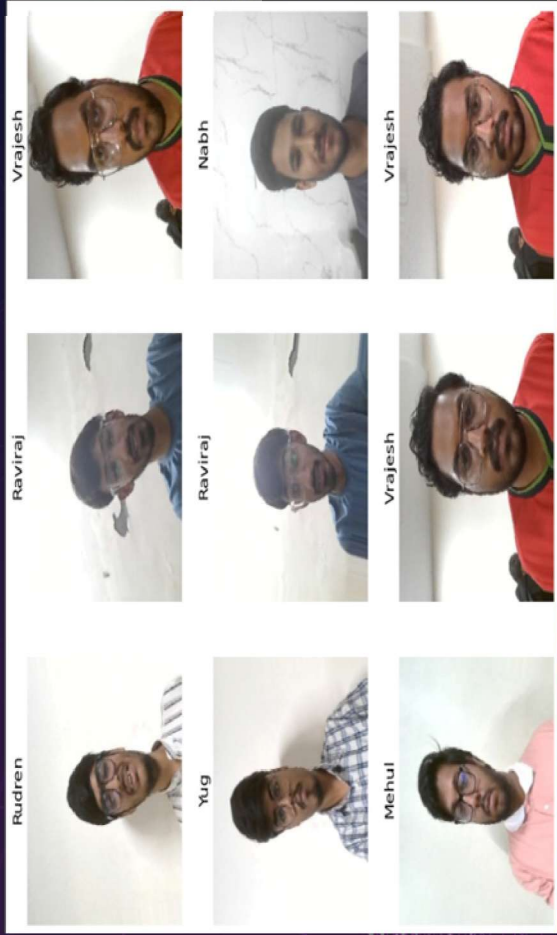
FACULTY MENTOR

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PROBLEM STATEMENT

While manual feature extraction in computer vision is limited by environmental volatility, this study explores the efficacy of hierarchical feature learning via CNNs and Transfer Learning. The problem lies in developing a model that can generalize across 12,000 images containing facial expression changes and perspective shifts. Our objective is to minimize misclassification through optimized architectures like MobileNetV2 and EfficientNet, establishing a benchmark for high-accuracy person identification.

DATASET OVERVIEW



Total	12,000	Classes	12
Train	9,600	Validation	2,400

MODEL ARCHITECTURE

Layer (type)	Output Shape	Param #
sequential (Sequential)	(None, 224, 224, 3)	0
conv2d (Conv2D)	(None, 224, 224, 32)	896
batch_normalization (BatchNormalization)	(None, 224, 224, 32)	128
activation (Activation)	(None, 224, 224, 32)	0
conv2d_1 (Conv2D)	(None, 224, 224, 32)	9,248
batch_normalization_1 (BatchNormalization)	(None, 224, 224, 32)	128
activation_1 (Activation)	(None, 224, 224, 32)	0
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)	0
dropout (Dropout)	(None, 112, 112, 32)	0
conv2d_2 (Conv2D)	(None, 112, 112, 64)	18,496
batch_normalization_2 (BatchNormalization)	(None, 112, 112, 64)	256
activation_2 (Activation)	(None, 112, 112, 64)	0
conv2d_3 (Conv2D)	(None, 112, 112, 64)	36,928
batch_normalization_3 (BatchNormalization)	(None, 112, 112, 64)	256
activation_3 (Activation)	(None, 112, 112, 64)	0
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)	0
dropout_1 (Dropout)	(None, 56, 56, 64)	0
conv2d_4 (Conv2D)	(None, 56, 56, 128)	73,856
batch_normalization_4 (BatchNormalization)	(None, 56, 56, 128)	512
activation_4 (Activation)	(None, 56, 56, 128)	0
conv2d_5 (Conv2D)	(None, 56, 56, 128)	147,584
batch_normalization_5 (BatchNormalization)	(None, 56, 56, 128)	512
activation_5 (Activation)	(None, 56, 56, 128)	0
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 128)	0
dropout_2 (Dropout)	(None, 28, 28, 128)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 128)	0
dense (Dense)	(None, 128)	16,512
batch_normalization_6 (BatchNormalization)	(None, 128)	512
dropout_3 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 12)	1,548

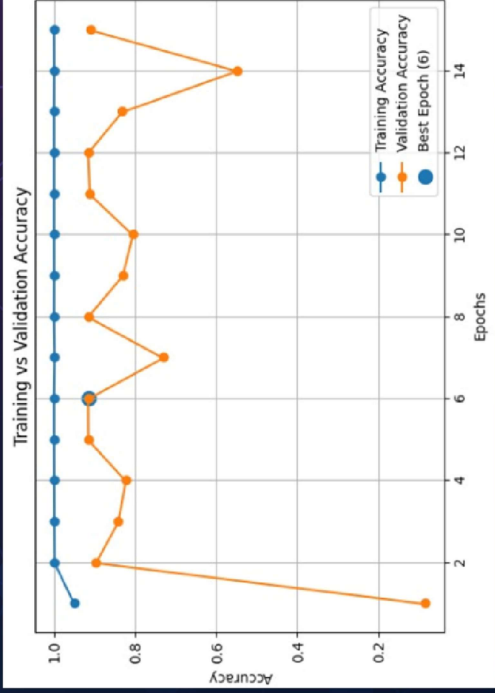
OPTIMIZATION & HYPERPARAMETERS

Optimizer	Adam
Learning Rate	0.0001
Dropout	0.25 - 0.5
Loss	Crossentropy
Dense Units	256

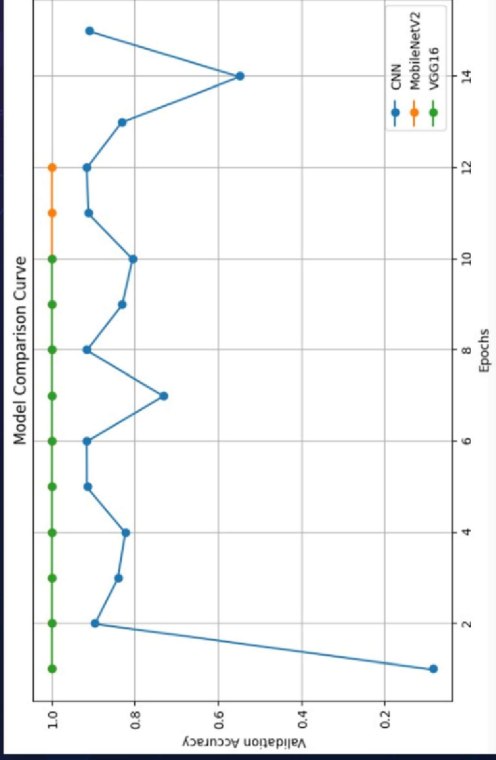
MODEL COMPARISON

Base CNN	95%
VGG16	99.8%
MobileNetV2	100%

TRAINING ACCURACY VS EPOCH



MODEL ACCURACY COMPARISON



BEST RESULT

100%
MobileNetV2

PREDICTION OUTPUT

